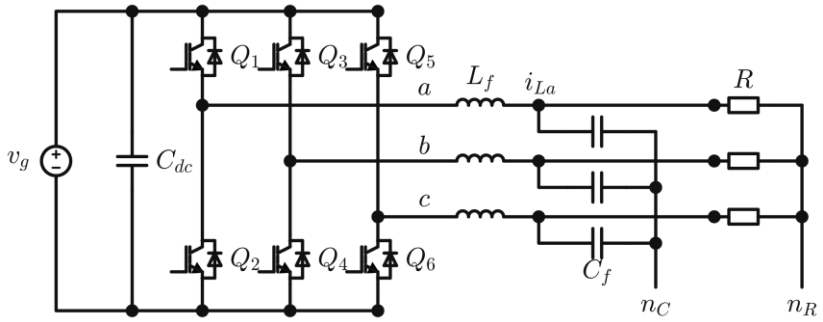
Project A Three phase standalone inverter

Build a three-phase inverter that supplies a balanced resistive load with regulated 60 Hz sinusoidal voltages.



v_g: ITECH source; Q1–Q6: SEMITEACH legs; L_f, C_f: filter branches; R: three-phase bank; n_C and n_R: capacitor and load star points.

Source	ITECH IT-M3632 bidirectional DC supply, 0–60 V, 30 A, 800 W; used in source mode with a 10 A current limit
Power stage	Three SEMITEACH legs; C_{dc} = 1.1 mF on the DC link
Filter	Per phase: one 47 µH ±20 % inductor and one 62 µF capacitor (LCL board wired as LC; second inductor unused). Capacitor star floating, 100 kΩ bleeders
Load	Three-phase resistor bank, 12.5–80 Ω per phase, balanced
Hardware limits	Switching ≤10 kHz. ITECH ≤58 V (OVP 60 V) and 10 A current limit (about 580 W at 58 V). IGBT 50 A. 47 µH inductor (Würth 7443783533470): 19.3 A rated current (50 K rise), 32.3 A typical saturation current (inductance −30 %).

What you decide and justify

- Operating points: at least two. For each, the output voltage (state phase or line-to-line), load resistance and output power.
- Switching frequency and control structure.
- Sensing: two differential voltage probes. Specify signals, probe settings and ADC scaling (Setup Guide); retain ADC headroom through trip levels.
- Protection: overcurrent and overvoltage thresholds and what the firmware does when they trip.
- The filter is fixed. Analyse how it and the load resistance affect your control loop.

Demonstration

1. Steady state at each operating point: two measured output line-to-line voltages and the reconstructed third, one inductor current, and DC input voltage and current from the ITECH display. Compare with predictions.
2. Load step: switch the bank between two settings you chose. Show two output line-to-line voltages and one inductor current through the transition.
3. Reference step: step the output-voltage amplitude reference between two values. Show the same signals. Compare both transients with your simulation.

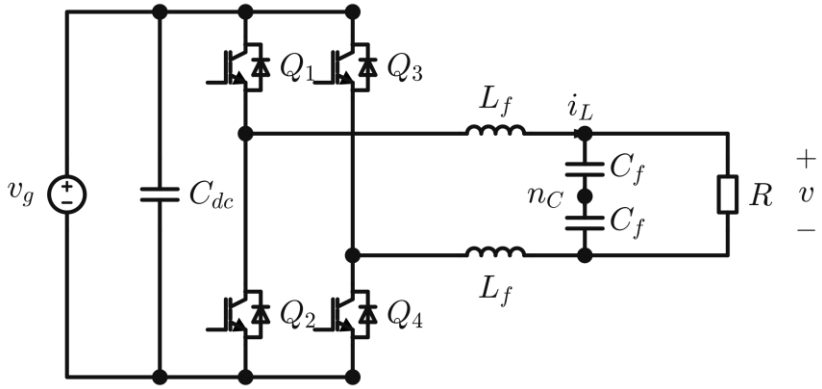
Station hazards

Keep n_C and n_R separate and floating; connect neither to earth nor DC−. Connect the load to the capacitor taps only, with unused inductor ends open. Leave the station transformer (used by C–E) disconnected from the inverter and load.

Common handout: deliverables (§1, App. A–E); bench sessions (§2); HIL check (App. C); requirements (App. G); precharge, startup and shutdown (App. H).

Project B Single phase standalone inverter

Build a single-phase inverter that supplies a resistive load with regulated 60 Hz sinusoidal voltage.



v_g: ITECH source; Q1–Q4: two SEMITEACH legs; L_f, C_f: two filter branches; R: bank across capacitor taps; n_C: capacitor star.

Source	ITECH IT-M3632 bidirectional DC supply, 0–60 V, 30 A, 800 W; used in source mode with a 10 A current limit
Power stage	Two SEMITEACH legs (third leg inhibited); C_{dc} = 1.1 mF on the DC link
Filter	Two branches of the LCL board, each one 47 µH ±20 % inductor and one 62 µF capacitor; load between the two capacitor taps. Capacitor star floating, 100 kΩ bleeders
Load	Single-phase resistor bank, 12.5–250 Ω (standard bank; see common handout for assigned alternatives)
Hardware limits	Switching ≤10 kHz. ITECH ≤58 V (OVP 60 V) and 10 A current limit (about 580 W at 58 V). IGBT 50 A. 47 µH inductor (Würth 7443783533470): 19.3 A rated current (50 K rise), 32.3 A typical saturation current (inductance −30 %).

What you decide and justify

- Operating points: at least two. For each, the output RMS voltage, load resistance and output power.
- Switching frequency, modulation and control structure.
- Sensing: two differential voltage probes. Specify signals, probe settings and ADC scaling (Setup Guide); retain ADC headroom through trip levels.
- Protection: overcurrent and overvoltage thresholds and what the firmware does when they trip.
- The filter is fixed. Analyse how it and the load resistance affect your control loop.

Demonstration

1. Steady state at each operating point: output voltage, inductor current, and the ITECH voltage and current readings. Compare with your predictions.
2. Load step: switch the bank between two settings you chose. Show the output voltage and inductor current through the transition.
3. Reference step: step the output-voltage amplitude reference between two values. Show the same signals. Compare both transients with your simulation.

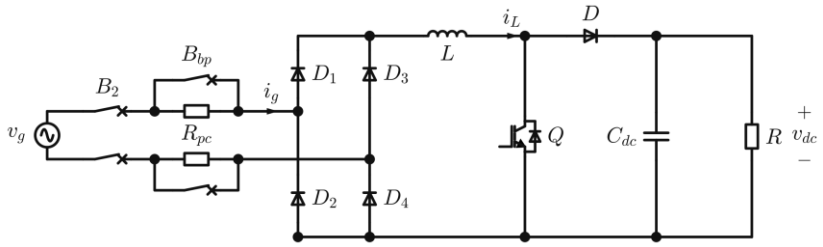
Station hazards

Keep n_C floating: no earth, neutral or DC− bond. Connect the load between the two capacitor taps only. Leave the third filter branch and unused inductor ends open. Leave the station transformer (used by C–E) disconnected from the inverter and load.

Common handout: deliverables (§1, App. A–E); bench sessions (§2); HIL check (App. C); requirements (App. G); precharge, startup and shutdown (App. H).

Project C Conventional boost PFC

Build a diode-bridge boost rectifier that regulates a DC output while drawing sinusoidal input current in phase with the transformer voltage.



v_g: transformer secondary; B_2, R_{pc}, B_{bp}: breaker and precharge; D1–D4: SEMITEACH rectifier; L: Hammond choke; Q: BOT1; D: TOP1 antiparallel diode (TOP1 held off); R: DC bank.

Source	Transformer secondary, line-to-line, 35–36.7 V RMS, 60 Hz (peak ≈52 V), on two of the three rectifier AC inputs; third input and neutral unused. Acuvim-L power meter on the transformer secondary
Power stage	Rectifier+ → L → leg 1 midpoint; rectifier− → DC−. Gate BOT1 only; TOP1 and all other devices inhibited. C_{dc} = 1.1 mF
Inductor	Hammond 195G20: 5 mH ±15 %, 0.025 Ω, 20 A DC rating
Load	Single-phase resistor bank across the DC link, 12.5–250 Ω (standard bank; see common handout for assigned alternatives)
Breakers and precharge	B_2: 20 A secondary breaker; closing it applies AC to your circuit. B_1 (not drawn): 16 A primary breaker; opening it de-energizes the transformer. R_{pc}: 30 Ω / 10 W in each line (60 Ω charging loop). B_{bp}: 16 A bypass breaker across R_{pc}; open while the DC link charges
Hardware limits	Switching ≤10 kHz. DC-voltage command ≤70 V. Inductor 20 A DC rating. IGBT 50 A. Bypass breaker 16 A (binds first). Transformer 400 VA per winding (about 19 A RMS)

What you decide and justify

• Operating points: at least two. For each, the DC output voltage (above the converter-side peak: transformer peak plus the inductor drop at your current), load resistance and output power.

• Switching frequency and control structure.

• Synchronization: build the input-current reference as a sinusoid from the transformer-voltage angle given by the provided example PLECS synchronization (PLL) model, not from a scaled copy of the measured voltage.

• Sensing: two differential voltage probes. Specify signals, probe settings and ADC scaling (Setup Guide); retain ADC headroom through trip levels.

• Protection: overcurrent and overvoltage thresholds and what the firmware does when they trip. Limit the DC-voltage command in firmware (≤70 V); trip below the DC voltage at which its ADC input saturates.

• Measurement: obtain true input PF and current THD from the meter/scope for this line-to-line connection; define phase error using the voltage and current fundamentals (common handout, App. G).

Demonstration

1. At each operating point: transformer voltage, input current and DC output voltage; calculate output current from Vdc/R. Report true PF, current THD and fundamental phase error. Compare with predictions.

2. Load step: switch the bank between two settings you chose. Show the DC voltage and input current through the transition.

3. Reference step: step the DC-voltage reference between two values. Show the same signals. Compare both transients with your simulation.

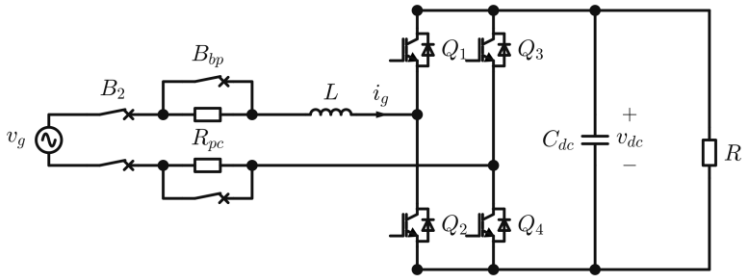
Station hazards

Never link rectifier+ directly to DC+. The rectifier and the TOP1 diode charge the link whenever AC is present, with all gates off. No ITECH on this station. Before closing B_2: B_{bp} open and DC load disconnected. Close B_{bp} once the DC-link voltage settles (common handout), then connect the load.

Common handout: deliverables (§1, App. A–E); bench sessions (§2); HIL check (App. C); requirements (App. G); precharge, startup and shutdown (App. H).

Project D Totem pole PFC

Build a bridgeless (totem-pole) boost rectifier that regulates a DC output while drawing sinusoidal input current in phase with the transformer voltage.



v_g: transformer secondary; B_2, R_{pc}, B_{bp}: breaker and precharge; L: Hammond choke; Q1/Q2: leg connected to the inductor; Q3/Q4: leg connected to the return; R: DC bank.

Source	Transformer secondary, line-to-line, 35–36.7 V RMS, 60 Hz (peak ≈52 V); neutral unused. Acuvim-L power meter on the transformer secondary
Power stage	Two SEMITEACH legs (third leg inhibited); C_{dc} = 1.1 mF
Inductor	Hammond 195G20: 5 mH ±15 %, 0.025 Ω, 20 A DC rating
Load	Single-phase resistor bank across the DC link, 12.5–250 Ω (standard bank; see common handout for assigned alternatives)
Breakers and precharge	B_2: 20 A secondary breaker; closing it applies AC to your circuit. B_1 (not drawn): 16 A primary breaker; opening it de-energizes the transformer. R_{pc}: 30 Ω / 10 W in each line (60 Ω charging loop). B_{bp}: 16 A bypass breaker across R_{pc}; open while the DC link charges
Hardware limits	Switching ≤10 kHz. DC-voltage command ≤70 V. Inductor 20 A DC rating. IGBT 50 A. Bypass breaker 16 A (binds first). Transformer 400 VA per winding (about 19 A RMS)

What you decide and justify

• Operating points: at least two. For each, the DC output voltage (above the converter-side peak: transformer peak plus the inductor drop at your current), load resistance and output power.

• Synchronization: build the input-current reference as a sinusoid from the transformer-voltage angle given by the provided example PLECS synchronization (PLL) model, not from a scaled copy of the measured voltage; use the angle to set which devices switch and conduct in each half cycle and the changeover at the zero crossings.

• Switching frequency and control structure.

• Sensing: two differential voltage probes. Specify signals, probe settings and ADC scaling (Setup Guide); retain ADC headroom through trip levels.

• Protection: overcurrent and overvoltage thresholds and what the firmware does when they trip. Limit the DC-voltage command in firmware (≤70 V); trip below the DC voltage at which its ADC input saturates.

• Measurement: obtain true input PF and current THD from the meter/scope for this line-to-line connection; define phase error using the voltage and current fundamentals (common handout, App. G).

Demonstration

1. At each operating point: transformer voltage, input current and DC output voltage; calculate output current from Vdc/R. Report true PF, current THD and fundamental phase error. Compare with predictions.

2. Load step between two bank settings you chose: show the DC voltage and input current through the step.

3. Reference step: step the DC-voltage reference between two values. Show the same signals. Compare both transients with your simulation.

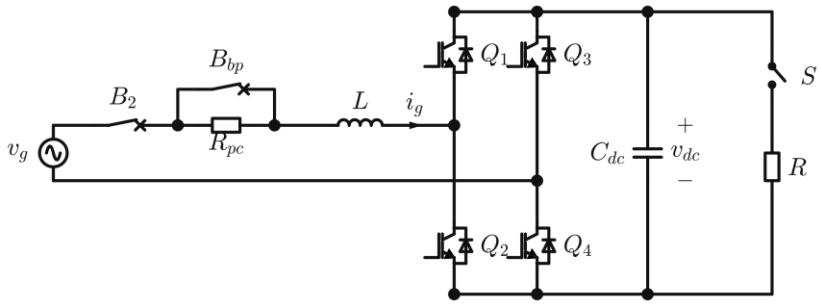
Station hazards

Before AC, verify gate assignments for both input polarities and prevent simultaneous turn-on in either leg. Diodes charge the link with gates off. No ITECH. Before closing B_2, open B_{bp} and disconnect the DC load. Close B_{bp} once the DC-link voltage settles (common handout), then connect the load.

Common handout: deliverables (§1, App. A–E); bench sessions (§2); HIL check (App. C); requirements (App. G); precharge, startup and shutdown (App. H).

Project E Single phase reactive power support

Build a grid-connected converter that regulates its own DC-link voltage and supplies a commanded reactive power to the transformer, with a resistor that can be switched across the DC link.



v_g: transformer secondary, phase-to-neutral; B_2, R_{pc}, B_{bp}: breaker and precharge; L: Hammond choke; Q1–Q4: two SEMITEACH legs; S: bank switch; R: DC bank.

Source	Transformer secondary, phase-to-neutral, 20.2–21.2 V RMS, 60 Hz (peak ≈30 V). Acuvim-L power meter on the transformer secondary
Power stage	Two SEMITEACH legs (third leg inhibited); C_{dc} = 1.1 mF
Inductor	Hammond 195G20: 5 mH ±15 %, 0.025 Ω, 20 A DC rating. No LCL board
Load	Single-phase resistor bank across the DC link, 12.5–250 Ω, switched by its own bank switches (standard bank; see common handout for assigned alternatives)
Breakers and precharge	B_2: 20 A secondary breaker; closing it applies AC to your circuit. B_1 (not drawn): 16 A primary breaker; opening it de-energizes the transformer. R_{pc}: one 30 Ω / 10 W resistor in the phase conductor; none in the neutral. B_{bp}: 16 A bypass breaker across R_{pc}; open while the DC link charges
Hardware limits	Switching ≤10 kHz. DC-voltage command ≤60 V. Inductor 20 A DC rating. IGBT 50 A. Bypass breaker 16 A (binds first). Transformer 400 VA per winding (about 19 A RMS)

What you decide and justify

• Operating points: DC-link voltage (above the converter-side peak: transformer peak plus the inductor drop at your current), inductive and capacitive reactive-power commands, DC resistor setting.

• Synchronization from the transformer-voltage angle of the provided example PLECS synchronization (PLL) model, current control and DC-voltage regulation. State your sign convention for real and reactive power.

• Switching frequency.

• Sensing: two differential voltage probes. Specify signals, probe settings and ADC scaling (Setup Guide); retain ADC headroom through trip levels.

• Protection: overcurrent and overvoltage thresholds and what the firmware does when they trip. Limit the DC-voltage command in firmware (≤60 V); trip below the DC voltage at which its ADC input saturates.

Demonstration

1. Steady state at one inductive and one capacitive Q command, resistor disconnected: transformer voltage and current, DC-link voltage. Compare measured Q with its command and measured P with your predicted losses.

2. Reference step: step the Q command from inductive to capacitive. Show the transformer current and DC-link voltage.

3. Load step: at one Q command, connect and then disconnect the DC resistor. Show the same signals. Compare all transients with your simulation.

Station hazards

Before energizing, measure phase-to-neutral voltage at the terminals you will use; never use line-to-line terminals. The antiparallel diodes charge the link whenever AC is present, with all gates off. No ITECH; LCL board disconnected. Before closing B_2: B_{bp} open and DC resistor disconnected. Close B_{bp} once the DC-link voltage settles.

Common handout: deliverables (§1, App. A–E); bench sessions (§2); HIL check (App. C); requirements (App. G); precharge, startup and shutdown (App. H).